Predicting Customer Subscription Status
Using Behavioral and Promotional Factors

A Manuscript
As a Partial Fulfillment of the
Final Requirements for the Course ISY122

By

Dumapias, Diane Rose T.
Haron, Usman

May 5, 2026

1. Abstract

This study examines whether customer subscription status can be predicted using behavioral and promotional factors in a Philippine coffee customer dataset. The analysis used 1,500 records with variables including review rating, previous purchases, product category, discount applied, and promo code usage. Three classification models were compared: Logistic Regression, Support Vector Machine (SVM), and Random Forest. The results showed that Random Forest achieved the best overall performance, indicating that machine learning can effectively classify subscription behavior in this context. Promotional variables were the strongest predictors, with discount application and promo code usage emerging as the most influential factors. Behavioral measures such as review ratings and previous purchases contributed less to prediction, although previous purchases may reflect non-linear patterns that machine learning methods capture more effectively. Overall, the findings suggest that customer subscription decisions are driven more by promotional incentives than by product category or basic behavioral indicators. These results support the use of predictive analytics to improve customer targeting and marketing strategy.

2. Keywords

Customer Subscription Prediction, Machine Learning Classification, Random Forest Algorithm, Behavioral Factors, Promotional Strategies

3. Introduction

In the competitive landscape of the Philippine coffee industry, businesses increasingly rely on data analytics to understand customer behavior and improve marketing decisions. This study addresses the need for structured predictive models that can classify customer subscription status based on behavioral and promotional variables. The selected independent variables include review rating, previous purchases, product category, discount applied, and promo code usage, which are considered relevant in influencing customer subscription behavior.

The study is grounded in consumer behavior and machine learning literature. Thaler's mental accounting framework explains how promotional cues such as discounts and promo codes can affect purchase decisions, while sales promotion research shows that price promotions influence perceived value and buying intentions (Thaler, 1985). Review ratings are also important because they function as quality signals and can shape consumer trust and purchasing decisions (Kutabish et al., 2023). By examining these factors together, the study aims to predict subscription status and identify which variables most strongly influence customer decisions.

4. Literature Review / Related Work

4.1 Promotional Factors and Consumer Decision-Making

Promotional strategies such as discounts and promo codes significantly influence consumer purchasing behavior. Thaler (1985) explains through mental accounting that consumers view discounts as gains rather than simple price reductions, increasing purchase likelihood, including subscription decisions. Supporting this, Chandon et al. (2000) found that sales promotions provide both functional and emotional benefits that reinforce buying behavior. Similarly, Shankar and Bolton (2004) noted that pricing strategies strongly affect consumer decisions and perceived value.

Recent studies confirm these effects, showing that promotional incentives remain key drivers of online purchasing behavior (Rolando, 2024) and are more effective when combined with digital engagement signals such as reviews and ratings (Zhang, Voorhees, & Hult, 2024).

4.2 Review Ratings and Social Influence

Online reviews and ratings serve as important indicators of product quality and trustworthiness. Cialdini (1984), through Social Proof Theory, explains that individuals rely on others' opinions when making decisions under uncertainty. High product ratings reduce perceived risk and increase the likelihood of purchase.

Recent literature strongly supports this influence. Wu (2024) found that online reviews significantly affect consumer purchasing decisions by shaping trust and perceived quality. Abell et al. (2024) further show that star ratings directly influence consumer reactions and decision confidence. In addition, Varga and Albuquerque (2023) highlight that negative reviews can significantly reduce purchase intention, reinforcing the importance of rating systems in consumer behavior.

4.3 Behavioral Factors and Purchase History

Customer behavior, particularly past purchasing activity, is widely used to predict future actions. Purchase frequency and prior transactions reflect customer engagement and can indicate readiness for subscription. Studies in marketing analytics suggest that historical purchase behavior is one of the strongest predictors of customer retention and future buying decisions, as it captures patterns of loyalty and consumption habits (Fader, 2012; V. Kumar & Reinartz, 2016).

4.4 Machine Learning in Customer Prediction

Machine learning techniques are widely used for predicting customer behavior due to their ability to model complex relationships among variables. Leo Breiman (2001) introduced the Random Forest algorithm, an ensemble learning method that improves prediction accuracy by combining multiple decision trees and reducing overfitting. Similarly, Cortes and Vapnik (1995) developed Support Vector Machines (SVM), which are effective for binary classification problems. Logistic Regression, as discussed by Hosmer et al. (2013), remains a widely used baseline model due to its interpretability. These models provide a strong foundation for predicting customer subscription status by integrating behavioral and promotional variables.

5. Research Questions

1. Can customer subscription status be predicted using behavioral and promotional factors?
2. What factors significantly influence customer subscription status?
3. Which classification model performs best in predicting customer subscription status?

6. Materials and Methods

6.1 SEMMA Framework

This study adopts the SEMMA (Sample, Explore, Modify, Model, Assess) framework as a structured approach to data mining and machine learning. SEMMA provides a clear and systematic process for handling data, from preparation to evaluation.

The framework ensures that the dataset is properly selected (Sample), patterns and relationships are understood (Explore), data is prepared for modeling (Modify), appropriate machine learning algorithms are applied (Model), and results are evaluated effectively (Assess).

Using SEMMA is important in this study as it ensures an organized and reliable workflow in predicting customer subscription status. It helps improve the accuracy of the models, supports better decision-making, and ensures that both behavioral and promotional factors are analyzed systematically.

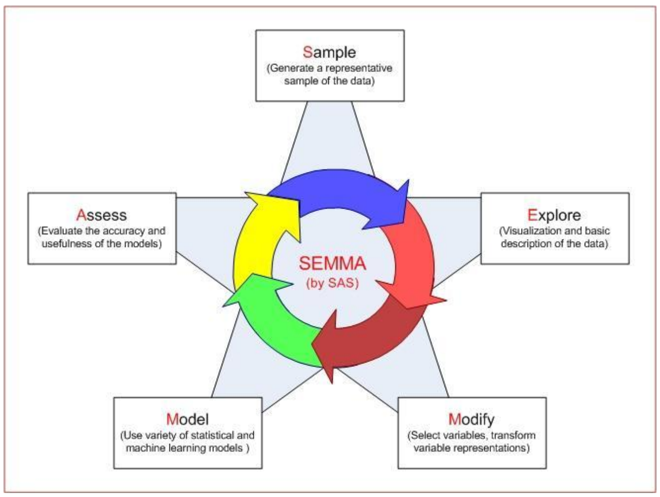


Figure 1. SEMMA Framework

6.1.1 Sample Phase

The dataset used in this study was obtained from the open-source Kaggle repository. It contains 1,500 rows and 14 columns representing customer purchasing behavior in the Philippines. The variables include age, gender, location, product category, season, review rating, previous purchases, payment method, purchase frequency, purchase amount, discount applied, promo code usage, and subscription status.

Based on initial inspection, the dataset contains no missing values, indicating that all records are complete and suitable for analysis. This structured dataset provides relevant behavioral and promotional information for predicting customer subscription decisions.

In this study, the dependent variable (Y) is subscription status, which indicates whether a customer subscribes to a service. The selected independent variables (X) include review rating, previous purchases, product category, discount applied, and promo code usage, as these variables are considered most relevant in influencing customer subscription behavior.

6.1.2 Explore Phase

Exploratory Data Analysis (EDA) was conducted to examine the distribution and relationships of the variables in the dataset. Visualization techniques such as bar charts and distribution plots were used to analyze patterns between customer attributes and subscription status.

6.1.3 Modify Phase

Data preprocessing was performed using Python in Google Colab to ensure that the dataset was suitable for analysis. The dataset was initially loaded, and only variables relevant to customer behavior and subscription status were selected for further processing. Categorical variables with Yes/No values were converted into numerical format (0 and 1) to ensure compatibility with analytical and machine learning processes. In addition, the Product Category variable was transformed using one-hot encoding to properly represent categorical data in numerical form without introducing ordinal relationships. The dataset was then organized by designating Subscription Status as the dependent (target) variable and placing it as the last column for clarity. The cleaned dataset was subsequently used for further analysis.

6.1.4 Model Phase

This study applied three classification models—Logistic Regression, Support Vector Machine (SVM), and Random Forest to predict customer subscription status. These models were selected to compare different learning approaches: Logistic Regression as a simple and interpretable baseline model, SVM for handling linear and non-linear class separation, and Random Forest as an ensemble method known for high accuracy and reduced overfitting in complex datasets.

Logistic Regression is widely used for binary classification due to its simplicity and interpretability (Hosmer et al., 2013). SVM is effective in high-dimensional classification problems and works well with clear margin separation (Cortes & Vapnik, 1995). Random Forest is a robust ensemble learning method that improves predictive performance by combining multiple decision trees (Breiman, 2001). The model with the highest accuracy was selected as the final predictive model.

1. Logistic Regression

Logistic Regression is a supervised machine learning algorithm commonly used for binary classification problems. It predicts the probability that a data instance belongs to a particular class by applying the logistic (sigmoid) function. In this study, it is used to classify whether a customer will subscribe ("Yes") or not ("No").

Formula:

$$P(Y=1)=\frac{1}{1+e^{-(\beta_0+\beta_1x_1+\beta_2x_2+\cdots+\beta_nx_n)}}$$

Where:

- $P(Y=1)$= probability of customer subscription
- e= Euler's number
- β_0= intercept
- $\beta_1,\beta_2,\ldots,\beta_n$= coefficients of predictors
- $x_1,x_2,\ldots,x_n$= independent variables

2. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised learning algorithm used for classification tasks. It works by identifying the optimal hyperplane that best separates classes while maximizing the margin between them. SVM is effective in handling both linear and non-linear classification problems.

Formula:

$$f(x)=w\cdot x+b$$

Where:

- w= weight vector

- x= input features
- b= bias term

Decision rule:

$$y = \begin{cases} 1, & \text{if } f(x) \geq 0 \\ 0, & \text{if } f(x) < 0 \end{cases}$$

3. **Random Forest**

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Each tree independently predicts the output, and the final classification is determined through majority voting.

Formula:

$$\hat{Y} = \text{majority vote}(T_1, T_2, T_3, \ldots, T_n)$$

Where:

- $\hat{Y}$= final predicted class
- $T_1, T_2, \ldots, T_n$= individual decision trees

6.1.4 Assess Phase

The final phase of the SEMMA framework involves evaluating the performance of the Logistic Regression, Support Vector Machine (SVM), and Random Forest models in predicting customer subscription status. Model performance was assessed using Accuracy, F-measure (F1-score), and Cohen's Kappa to determine the effectiveness, reliability, and classification capability of each model.

1. **Accuracy**

Accuracy measures the overall proportion of correctly classified instances among all predictions made by the model.

Formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP= True Positives
- TN= True Negatives
- FP= False Positives
- FN= False Negatives

2. **F-measure (F1-Score)**

The F-measure evaluates the balance between Precision and Recall. It is useful when assessing how well the model identifies subscribed customers.

Formula:

$$F1 = \frac{2 \times (Precision \times Recall)}{Precision + Recall}$$

Where:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

3. **Cohen's Kappa**

Cohen's Kappa measures the agreement between predicted and actual classifications while considering the possibility of agreement occurring by chance.

Formula:

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

Where:

- κ= Cohen's Kappa
- P_o= observed agreement
- P_e= expected agreement by chance

A higher Kappa value indicates stronger model reliability and classification agreement.

6.2 Simulation Workflow

To demonstrate the practical application of the study, a predictive simulation was integrated into the KNIME Analytics Platform to test the model's performance on new, hypothetical data. This process uses a Table Creator node to manually enter specific customer profiles, which are then processed by a One-to-Many encoding node to ensure the data structure remains consistent with the original training set. These inputs are fed into a secondary Predictor node that applies the logic from the previously trained algorithm to generate a real-time classification. The model that achieved the highest overall accuracy and predictive reliability during the assessment phase was selected as the engine for this simulation.

7. Results

7.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to examine and understand the patterns and relationships within the customer dataset. This phase focuses on analyzing how different customer behavior variables relate to subscription status, including product category, review rating, discount usage, promo code usage, and previous purchases.

1. Subscription Status by Coffee Category

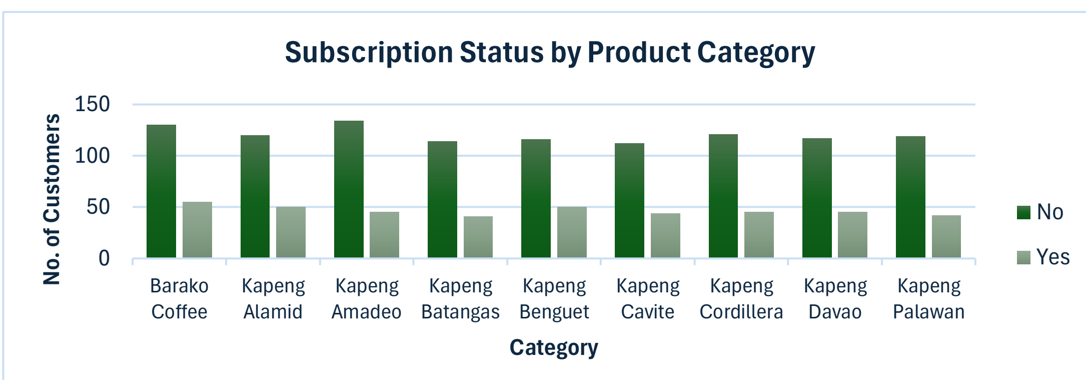


Figure 2. Subscription Status by Coffee Category

It shows that non-subscribed customers consistently outnumber subscribed customers across all coffee types, indicating a low overall subscription rate. While some categories, such as Barako Coffee, show slightly higher subscriber numbers, the differences across coffee types are minimal. Overall, the distribution of subscription status appears relatively uniform.

2. Customer Review Rating by Subscription Status

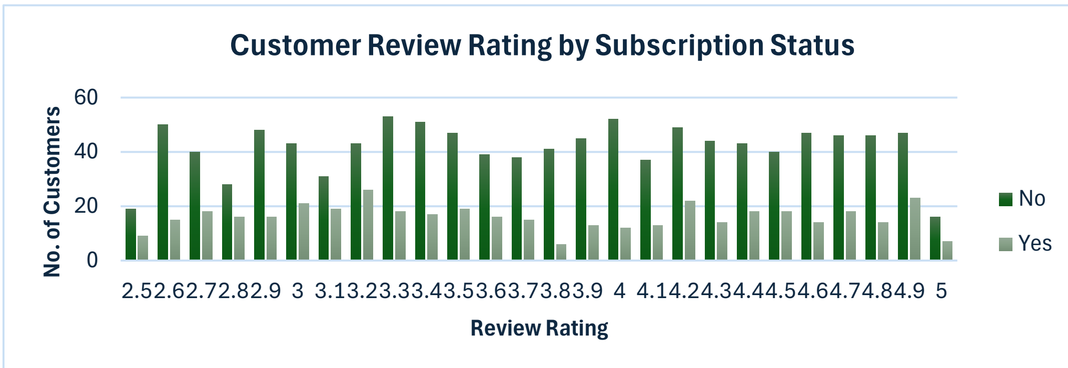


Figure 3. Customer Review Rating by Subscription Status

It shows that non-subscribed customers consistently outnumber subscribed customers across all review rating levels, indicating that subscription remains lower regardless of rating. While there are slight increases in the number of subscribed customers at certain mid-to-high rating levels (such as around 3.2, 4.2, and 4.9), the pattern is not consistent across all ratings. Overall, both subscribed and non-subscribed customers are distributed across the full range of ratings.

3. Subscription Status by Discount Application

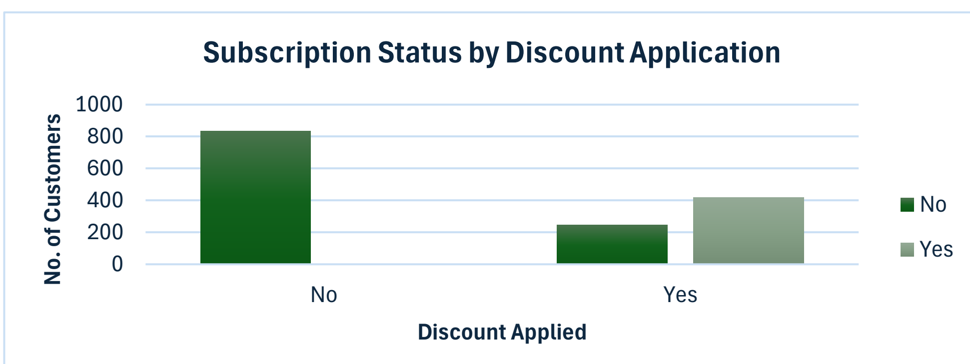


Figure 4. Subscription Status by Discount Application

The graph shows that among non-subscribed customers, the majority did not receive a discount, with 835 cases compared to 248 who received one. In contrast, all subscribed customers are associated with cases where a discount was applied, totaling 417. This indicates a noticeable difference in how discounts are distributed between the two groups, with discounts appearing more commonly among subscribed customers.

4. Subscription Status by Promo Code Usage

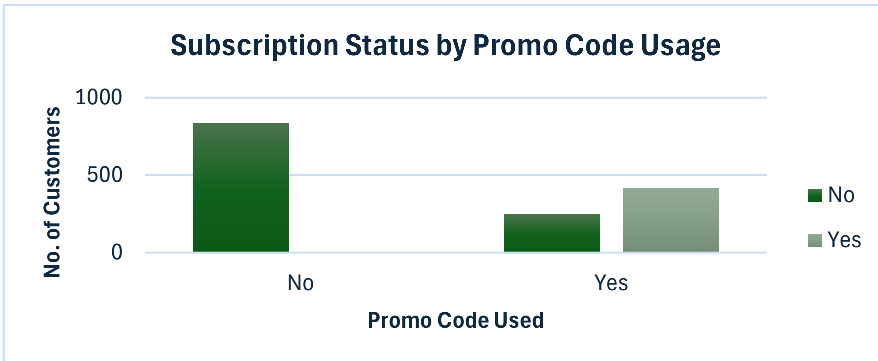


Figure 5. Subscription Status by Promo Code Usage

The graph shows that among non-subscribed customers, the majority did not use a promo code, with 835 cases compared to 248 who used one. In contrast, all subscribed customers are associated with cases where a promo code was used, totaling 417. This highlights a clear difference between the two groups, where promo code usage is more commonly observed among subscribed customers.

5. Previous Purchases by Subscription Status

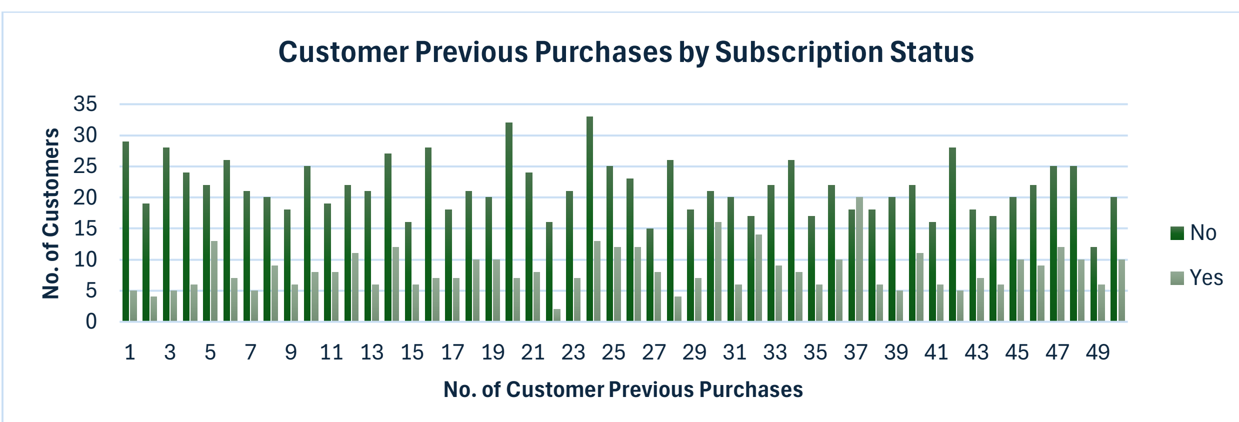


Figure 6. Customer Previous Purchases by Subscription Status

The graph shows that non-subscribed customers consistently outnumber subscribed customers across all levels of previous purchases. However, subscribed customers are distributed across a wide range of purchase frequencies, with slightly higher counts appearing at certain higher purchase levels (such as around 30, 32, 37, and 40). While there are some fluctuations in the number of subscribed customers as purchase frequency increases, both groups are present across low to high purchase counts, indicating varied purchasing behavior among customers.

6. Subscription Status Distribution of Customers

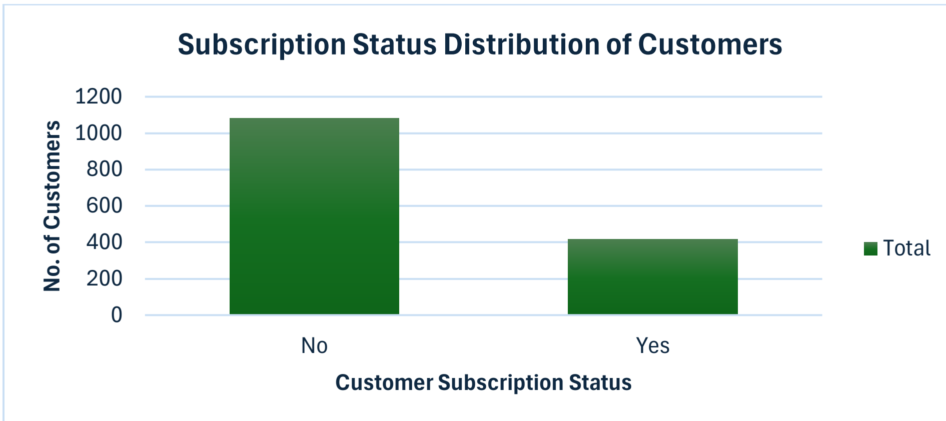


Figure 7. Subscription Status Distribution of Customers

The graph shows that the majority of customers are not subscribed, with 1,083 cases compared to 417 subscribed customers out of a total of 1,500. This indicates that non-subscribed customers make up a significantly larger portion of the dataset, reflecting a lower overall subscription rate among customers.

Correlation and Predictive Validation

Table 1. Rule of Thumb for Pearson's Correlation Coefficient

Correlation Coefficient (r)	Strength of Relationship
0.00 – 0.19	Very Weak
0.20 – 0.39	Weak
0.40 – 0.59	Moderate
0.60 – 0.79	Strong
0.80 – 1.00	Very Strong

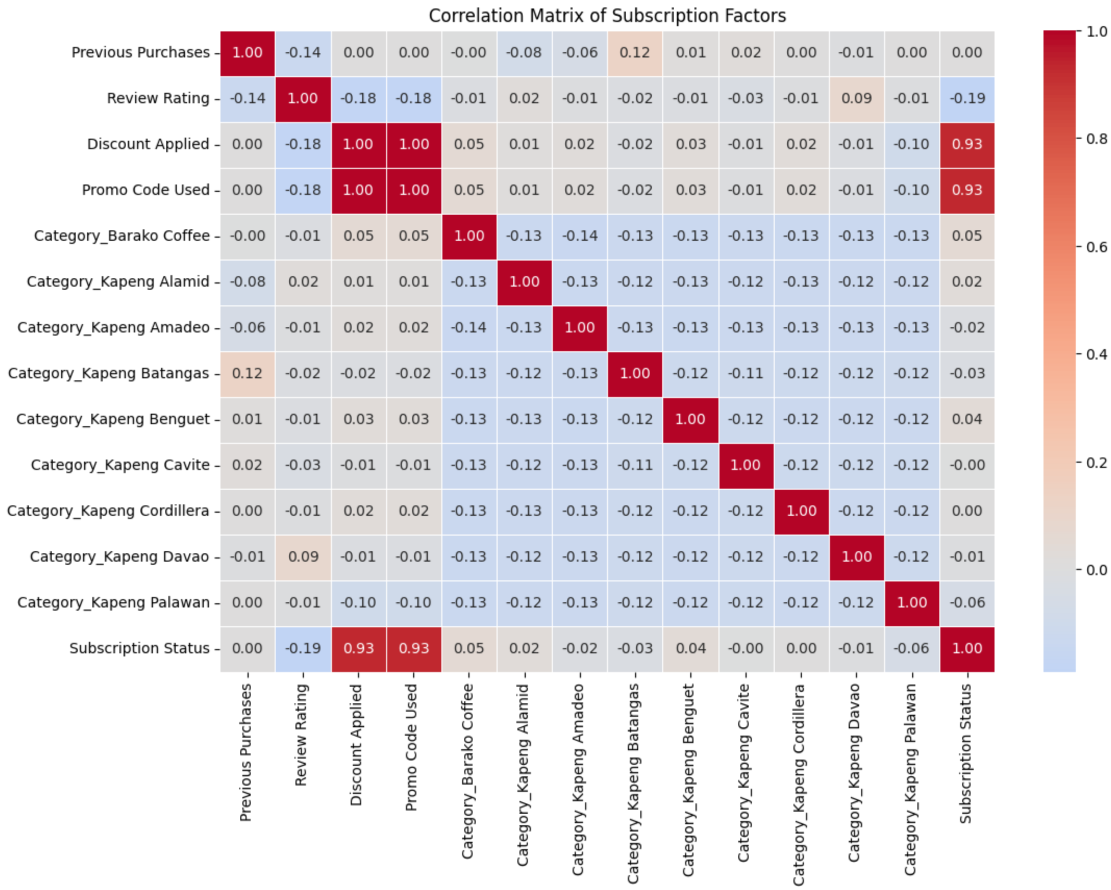


Figure 8. Correlation Matrix of Customer Subscription Factors

Figure 8 reveals that subscription status is primarily driven by promotional factors, evidenced by the very strong positive relationship (r = 0.93) with both discount applied and promo code used. The 1.00 correlation between these two predictors indicates perfect multicollinearity, meaning they are statistically identical and provide redundant information to the model. Interestingly, previous purchases showed a neutral linear correlation (r = 0.00) yet ranked high in Feature Importance; this suggests the Random Forest captured a non-linear behavioral pattern (e.g., a specific frequency of purchase) that linear correlation coefficients cannot detect. Conversely, all other factors including review ratings (-0.19), and all Coffee Categories (0.00 to 0.06) show very weak correlations with the target. This confirms that the decision to subscribe is triggered almost exclusively by price reductions rather than product variety or customer history.

7.3 Model Performance

The table below summarizes the key performance metrics for Logistic Regression, Support Vector Machine (SVM), and Random Forest. The models were evaluated based on their Overall Accuracy, F-measure (for the "Yes" class), and Cohen's Kappa.

Table 1. Model Accuracy

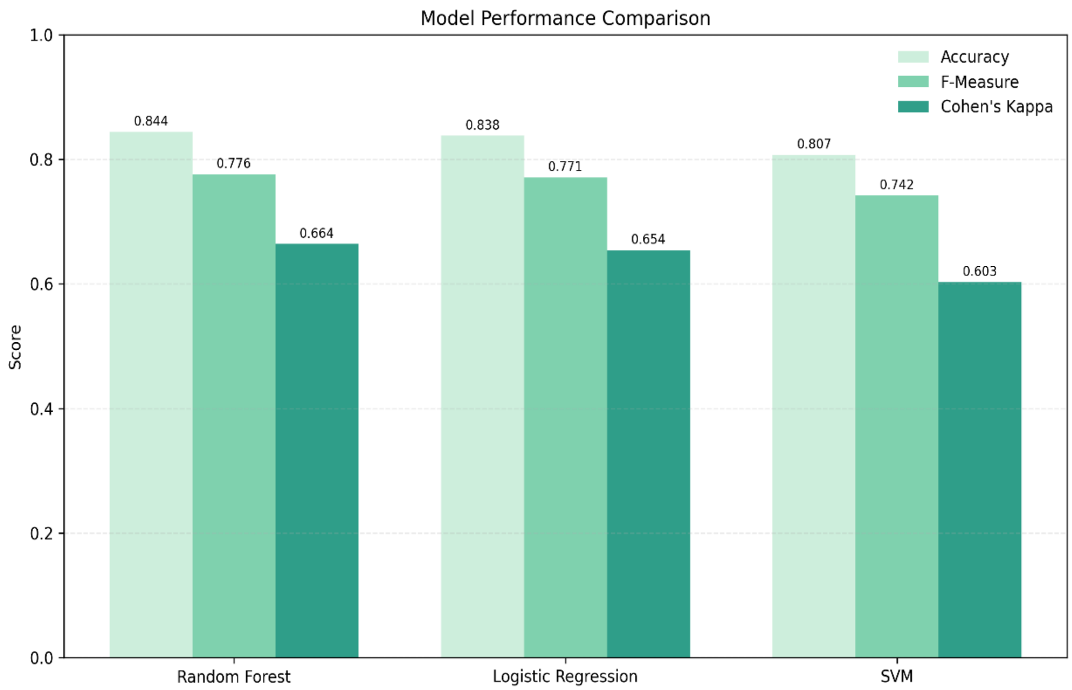


Figure 9. Model Performance Comparison

The figure indicates that the Random Forest model achieved the highest overall performance (84.4%), demonstrating substantial agreement according to its Kappa value and showing high sensitivity in identifying subscribers. Logistic Regression served as a strong baseline model, maintaining predictive stability and scores (83.8%) that were very close to the Random Forest. Meanwhile, the Support Vector Machine (SVM) provided good overall accuracy (80.7%) but lagged the other two models due to a notably lower recall for the subscriber segment. Based on these findings, the Random Forest model was selected as the final predictive engine because it best balanced the need to capture potential subscribers while maintaining the highest overall accuracy.

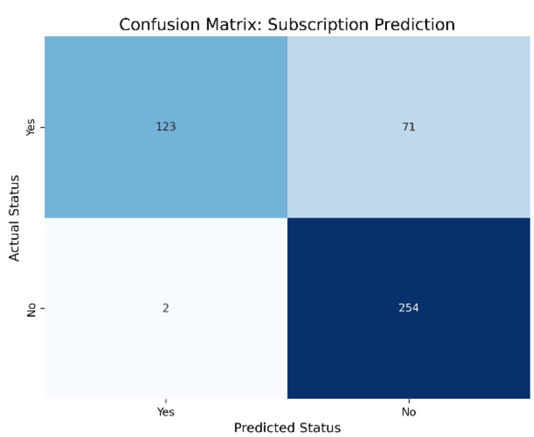
7.4 Model Interpretation

Interpretation of the model focused on Feature Importance, which quantifies the relative contribution of each predictor to the final forecast. Using the Gini Importance principle (Breiman, 2001), variables were ranked by their predictive power based on Level 0 splits—the primary decision points in the model.

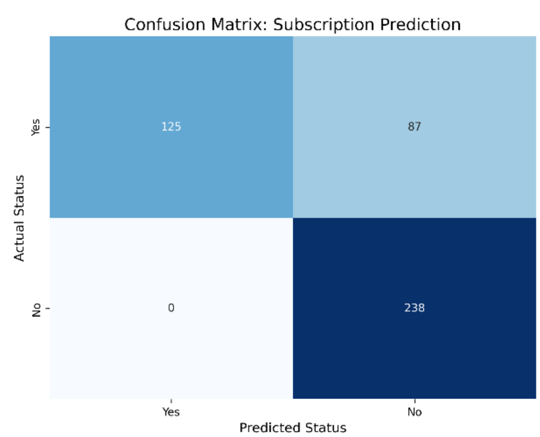
Table 2. Feature Importance Ranking

Feature	Splits at Level 0	Interpretation
Discount	128	Primary Predictor. The model relies on discounts more than any other variable.
Promo Code	94	High-Influence. Promotional codes are a critical secondary driver for subscriptions.
Previous	81	Strong Behavioral Driver. History is a reliable indicator of future intent.
Review	66	Moderate Influencer. Ratings provide context but are not as decisive as promotions.
Category	0 to 45	Marginal Factor. Coffee type has the least impact on the decision to subscribe.

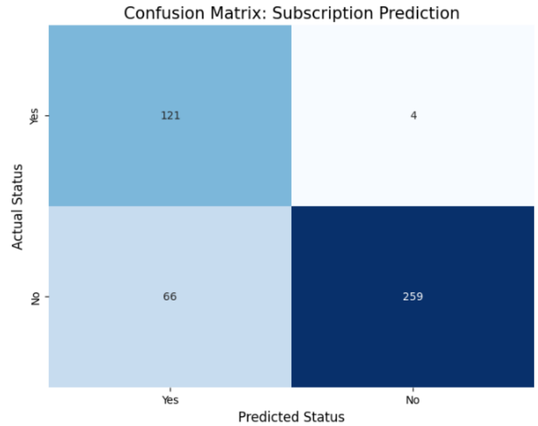
7.5 Confusion Matrix



Logistic Regression



Support Vector Machine



Random Forest

Figure 9. Confusion Matrix of Customer Subscription Prediction

The confusion matrices present the classification performance of the three models in predicting customer subscription status.

For ***Logistic Regression***, the model correctly classified 123 subscribers (True Positives) and 254 non-subscribers (True Negatives). It produced 2 False Negatives, indicating a strong ability to detect actual subscribers, but recorded 71 False Positives, suggesting a tendency to overestimate subscription outcomes.

The ***Support Vector Machine (SVM)*** model correctly identified 125 subscribers and 238 non-subscribers. Notably, it yielded 0 False Negatives, demonstrating perfect sensitivity in detecting actual subscribers. However, it generated 87 False Positives, indicating the highest rate of over-prediction among the three models.

In comparison, the **Random Forest** model correctly classified 121 subscribers and 259 non-subscribers. It produced 4 False Negatives and 66 False Positives, reflecting a more balanced distribution of errors. While it does not completely eliminate missed subscribers, it significantly reduces incorrect positive predictions relative to the other models.

Overall, the Random Forest model demonstrates the most optimal performance, achieving the highest accuracy and a better balance between sensitivity and specificity. This makes it the most suitable model for final customer subscription prediction in the study.

7.6 Predictive Simulation on New Dataset

1: Prediction output | Flow Variables

Rows: 50 | Columns: 14

Table | Statistics

RowID	Previous ... Number (I...	Review R... Number (...	Discount ... Number (I...	Promo Co... Number (I...	Category... Number (I...	Category... Number (I...	Category... Number (I...	Category... Number (I...	Category... Number (I...	Category... Number (I...	Category... Number (I...	Category... Number (I...	Category... Number (I...	Predictio... String
Row0	10	2.1	1	0	0	0	0	1	0	0	0	0	0	No
Row1	5	7	0	0	1	0	0	0	0	0	0	0	0	No
Row2	10	10	1	1	0	1	0	0	0	0	0	0	0	Yes
Row3	1	2	1	1	0	1	0	0	0	0	0	0	0	Yes

Figure 10. Prediction Output on New Dataset

This presents the prediction output for the new dataset using the Random Forest model. Each row represents a simulated customer profile, and the final column shows the predicted subscription status ("Yes" or "No"). The results indicate that the model can effectively classify new inputs, demonstrating its ability to generalize and support decision-making for customer subscription prediction.

8. Discussion and Conclusion

8.1 Interpretation of Findings

This section interprets the findings of the study based on the three research questions.

RQ 1: Can customer subscription status be predicted using behavioral and promotional factors?

The results indicate that customer subscription status can be reliably predicted using a combination of behavioral and promotional variables. All three models—Logistic Regression, Support Vector Machine (SVM), and Random Forest—achieved strong predictive performance, with accuracies ranging from 80.7% to 84.4%. The Random Forest model demonstrated the best overall performance, as reflected in its higher accuracy, F-measure, and Cohen's Kappa, indicating more balanced and reliable classification. The results from the confusion matrix further confirm that the model effectively distinguishes between subscribers and non-subscribers. Overall, this suggests that the dataset contains meaningful patterns that allow subscription behavior to be predicted with a high degree of reliability.

RQ 2: What factors significantly influence customer subscription status?

The findings show that promotional factors are the most influential determinants of customer subscription behavior. Specifically, discount application and promo code usage consistently emerged as the strongest predictors, supported by both correlation results and feature importance rankings. This indicates that subscription decisions are largely driven by financial incentives rather than product-related or demographic characteristics. In contrast, behavioral variables such as review ratings and previous purchases exhibited weaker linear relationships with subscription status. However, previous purchases still contributed to model performance, suggesting that its influence may be non-linear and only captured through machine learning methods. Product category showed minimal predictive value, indicating limited influence on subscription decisions.

RQ 3: Which classification model performs best in predicting customer subscription status?

Among the models evaluated, Random Forest performed best in predicting customer subscription status. It consistently outperformed Logistic Regression and SVM in terms of overall accuracy and class balance, particularly in identifying subscribers. Logistic Regression provided stable and competitive results as a baseline model, while SVM showed comparatively lower performance in capturing complex patterns within the dataset. These findings suggest that ensemble-based methods are more effective for this problem due to their ability to model non-linear relationships and interactions among variables.

Thus, the findings demonstrate that customer subscription behavior is primarily driven by promotional incentives, with discounts and promo codes serving as the most decisive factors. Behavioral characteristics play a secondary role, while product-related attributes have minimal impact. The results also highlight the effectiveness of machine learning—particularly ensemble methods in capturing complex behavioral patterns for subscription prediction.

8.2 Conclusion

The study concludes that customer subscription status can be accurately predicted using behavioral and promotional factors, with Random Forest performing best among the tested models. Promotional factors, particularly discount application and promo code usage, are the strongest influences on subscription decisions, while behavioral factors have weaker effects, and product category has minimal impact. Overall, subscription behavior is mainly driven by promotional incentives, and ensemble machine learning models are effective in capturing these patterns.

8.3 Limitations

This study is limited by the use of a single dataset sourced from Kaggle, which may not fully represent broader customer populations or real-world subscription behavior. The analysis is also restricted to a selected set of behavioral and promotional variables, meaning other potentially relevant factors such as income level, customer satisfaction, or marketing exposure were not included. In addition, the dataset is cross-sectional, which prevents analysis of changes in customer behavior over time. Lastly, while the models performed well, their results are dependent on the quality and structure of the dataset, and may vary when applied to different contexts or industries.

8.4 Future Work

Future studies may improve prediction accuracy by using larger and more diverse datasets that better represent different customer segments and industries. Additional variables such as income level, customer satisfaction, and marketing exposure can also be included to provide a more comprehensive analysis of subscription behavior. Future research may also explore advanced machine learning models such as gradient boosting methods or neural networks to compare performance with Random Forest. In addition, applying time-based or longitudinal data could help capture changes in customer behavior over time and improve the understanding of subscription patterns.

References

Abell, A., Morgan, C., & Romero, M. (2024). The power of a star rating: Effects on consumer reactions. *Journal of Marketing Research*. https://doi.org/10.1177/00222437241240694

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Chandon, P., Wansink, B., & Laurent, G. (2000). A benefit congruency framework of sales promotion effectiveness. *Journal of Marketing, 64*(4), 65–81. https://doi.org/10.1509/jmkg.64.4.65.18071

Cialdini, R. (1984). Influence. The Psychology of Persuasion. New York, NY: William Morrow e Company.

Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning, 20*(3), 273–297. https://doi.org/10.1007/BF00994018

Fader, P. S. (2012). *Customer Centricity: Focus on the Right Customers for Strategic Advantage*. Wharton Digital Press.

Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression* (3rd ed.). Wiley.

Kumar, V., & Reinartz, W. (2016). Creating enduring customer value. Journal of Marketing, 80(6), 36–68. https://doi.org/10.1509/jm.15.0414

Kutabish, S., Soares, A. M., & Casais, B. (2023). The influence of online ratings and reviews in consumer buying behavior: A systematic literature review. In *International Conference on Digital Economy*. https://doi.org/10.1007/978-3-031-42788-6_8

Rolando, B. (2024). Pricing strategies and their effects on online purchasing behavior: A systematic literature review. *International Journal of Economics and Business Studies*.

Shankar, V., & Bolton, R. N. (2004). An empirical analysis of determinants of retailer pricing strategy. *Marketing Science, 23*(1), 28–49. https://doi.org/10.1287/mksc.1030.0034

Thaler, R. (1985). Mental accounting and consumer choice. *Marketing Science, 4*(3), 199–214. https://doi.org/10.1287/mksc.4.3.199

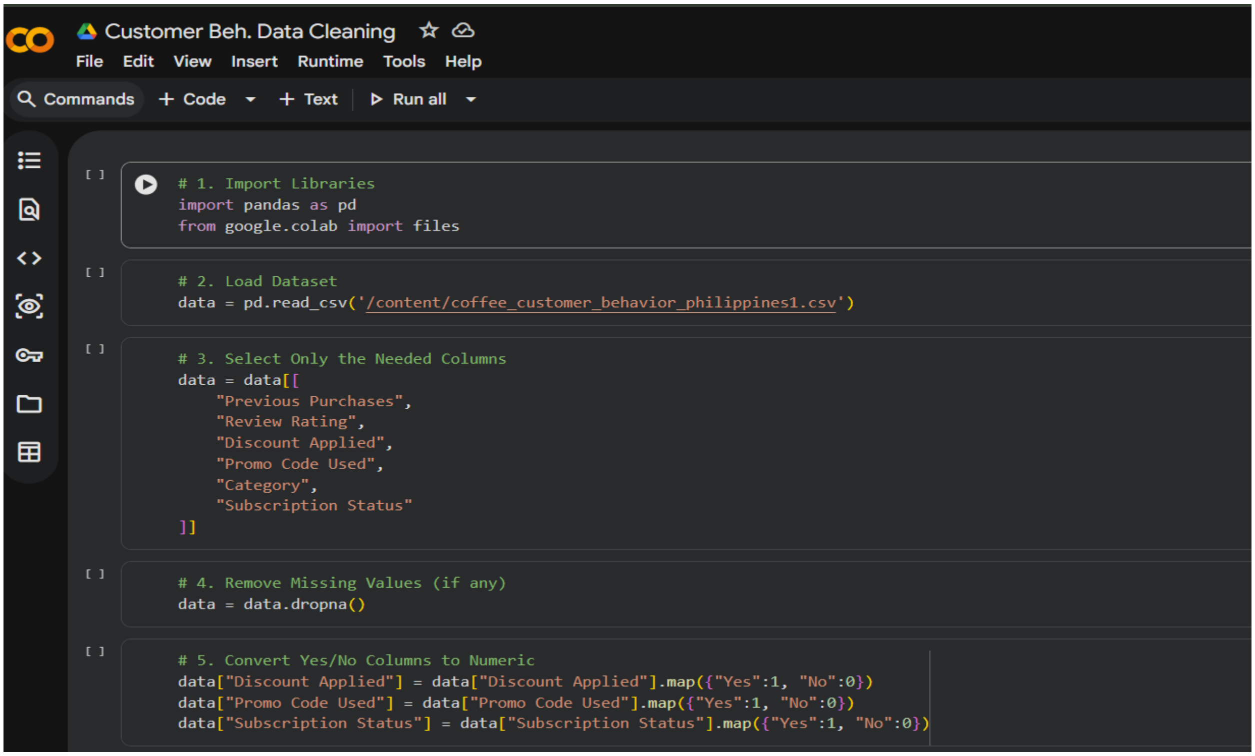
Varga, M., & Albuquerque, P. (2023). The impact of negative reviews on purchase decisions. *Journal of Marketing Research*. https://doi.org/10.1177/00222437231190874

Wu, Q.-M. (2024). The influence of online reviews on the purchasing decisions of consumers. *Sustainability, 16*(8), 3213. https://www.mdpi.com/2071-1050/16/8/3213

Zhang, Y., Voorhees, C. M., & Hult, G. T. M. (2024). Dynamic interplays between online reviews and marketing promotions. *Journal of the Academy of Marketing Science*. https://doi.org/10.1007/s11747-024-01025-w

Appendix

a. Code for Data Preprocessing in Google Colab



Customer Beh. Data Cleaning

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

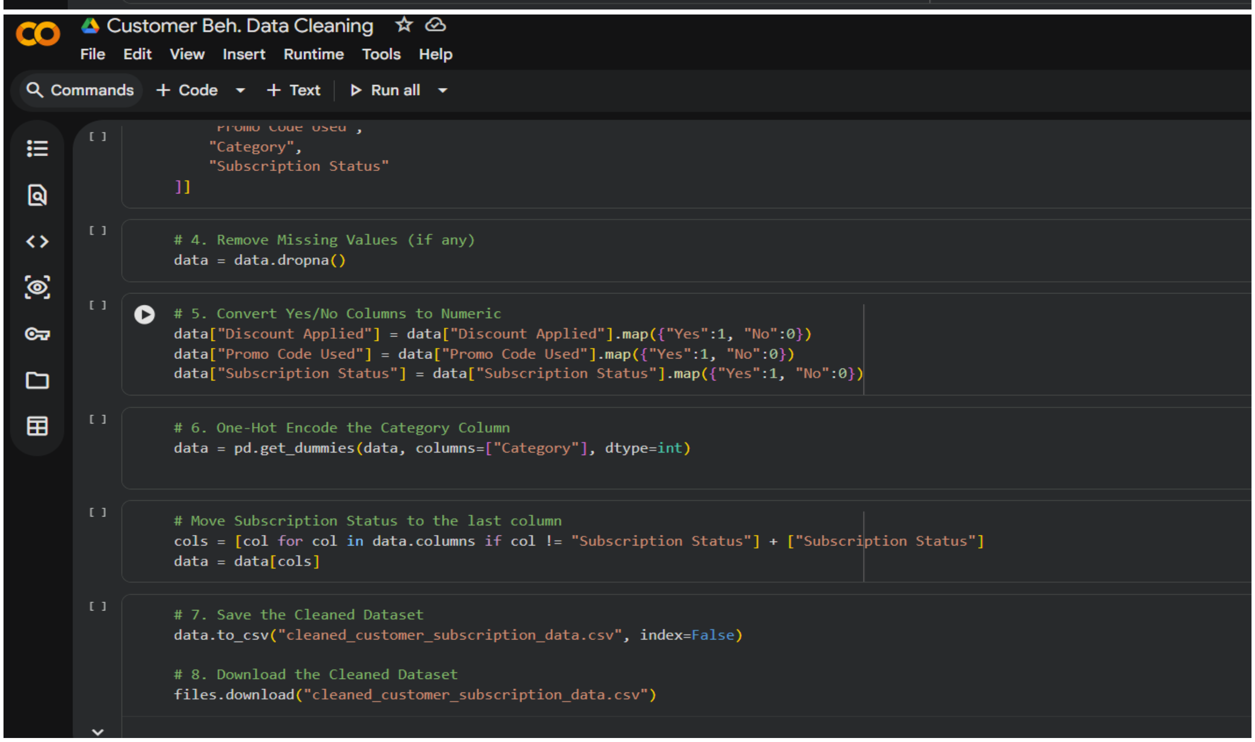
```python
# 1. Import Libraries
import pandas as pd
from google.colab import files
```

```python
# 2. Load Dataset
data = pd.read_csv('/content/coffee_customer_behavior_philippines1.csv')
```

```python
# 3. Select Only the Needed Columns
data = data[[
    "Previous Purchases",
    "Review Rating",
    "Discount Applied",
    "Promo Code Used",
    "Category",
    "Subscription Status"
]]
```

```python
# 4. Remove Missing Values (if any)
data = data.dropna()
```

```python
# 5. Convert Yes/No Columns to Numeric
data["Discount Applied"] = data["Discount Applied"].map({"Yes":1, "No":0})
data["Promo Code Used"] = data["Promo Code Used"].map({"Yes":1, "No":0})
data["Subscription Status"] = data["Subscription Status"].map({"Yes":1, "No":0})
```



Customer Beh. Data Cleaning

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```python
    "Promo Code Used",
    "Category",
    "Subscription Status"
]]
```

```python
# 4. Remove Missing Values (if any)
data = data.dropna()
```

```python
# 5. Convert Yes/No Columns to Numeric
data["Discount Applied"] = data["Discount Applied"].map({"Yes":1, "No":0})
data["Promo Code Used"] = data["Promo Code Used"].map({"Yes":1, "No":0})
data["Subscription Status"] = data["Subscription Status"].map({"Yes":1, "No":0})
```

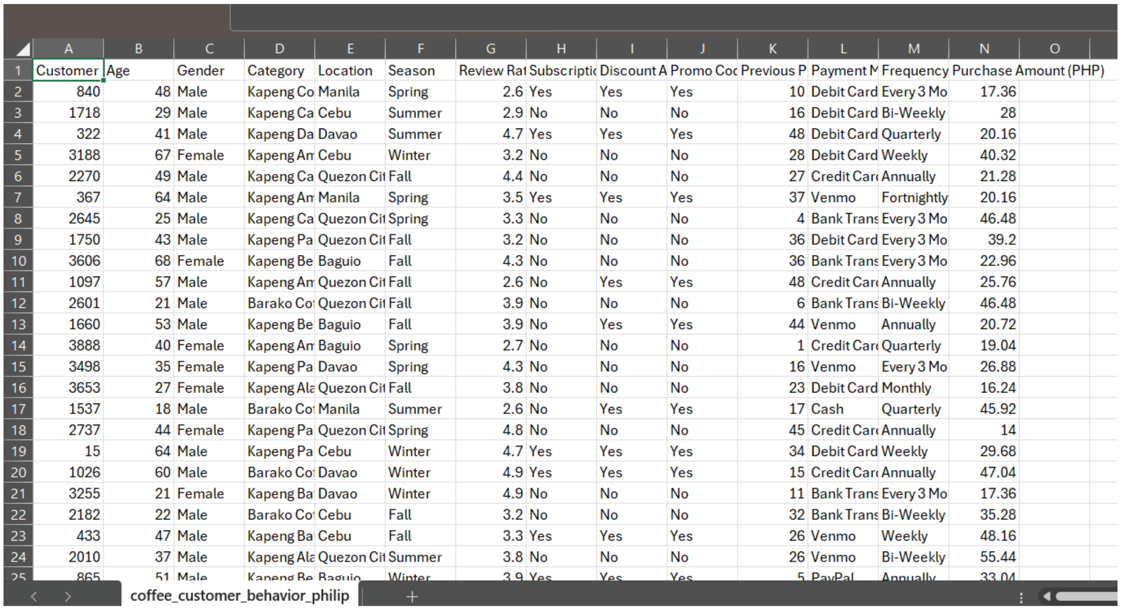
```python
# 6. One-Hot Encode the Category Column
data = pd.get_dummies(data, columns=["Category"], dtype=int)
```

```python
# Move Subscription Status to the last column
cols = [col for col in data.columns if col != "Subscription Status"] + ["Subscription Status"]
data = data[cols]
```

```python
# 7. Save the Cleaned Dataset
data.to_csv("cleaned_customer_subscription_data.csv", index=False)

# 8. Download the Cleaned Dataset
files.download("cleaned_customer_subscription_data.csv")
```

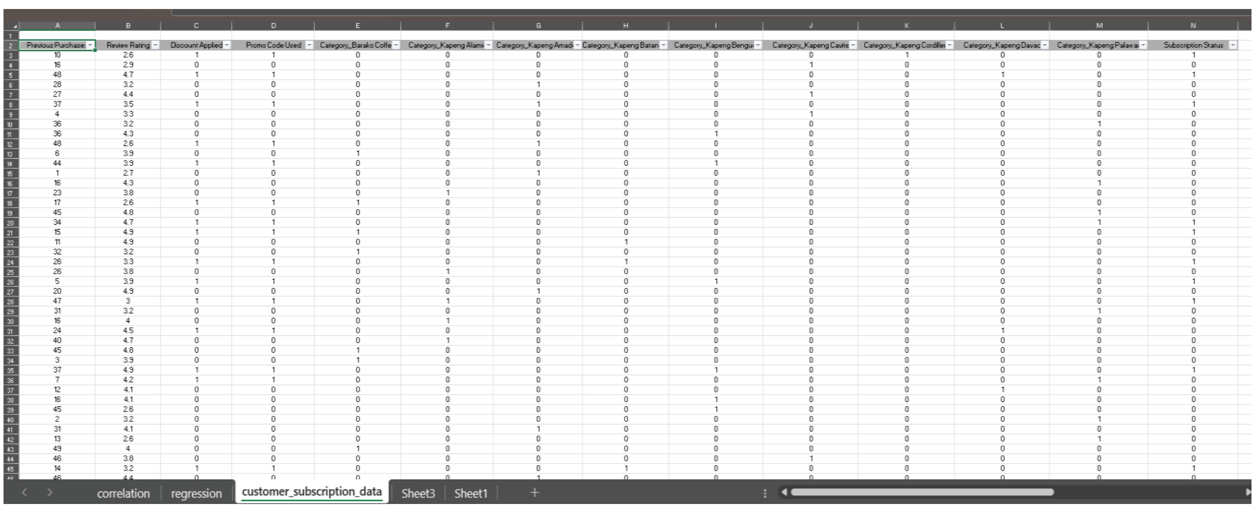
b. Raw Data



Customer	Age	Gender	Category	Location	Season	Review Rat	Subscriptic	Discount A	Promo Coc	Previous P	Payment M	Frequency	Purchase Amount (PHP)
840	48	Male	Kapeng Co	Manila	Spring	2.6	Yes	Yes	Yes	10	Debit Card	Every 3 Mo	17.36
1718	29	Male	Kapeng Ca	Cebu	Summer	2.9	No	No	No	16	Debit Card	Bi-Weekly	28
322	41	Male	Kapeng Da	Davao	Summer	4.7	Yes	Yes	Yes	48	Debit Card	Quarterly	20.16
3188	67	Female	Kapeng Am	Cebu	Winter	3.2	No	No	No	28	Debit Card	Weekly	40.32
2270	49	Male	Kapeng Ca	Quezon Cit	Fall	4.4	No	No	No	27	Credit Card	Annually	21.28
367	64	Male	Kapeng Am	Manila	Spring	3.5	Yes	Yes	Yes	37	Venmo	Fortnightly	20.16
2645	25	Male	Kapeng Ca	Quezon Cit	Spring	3.3	No	No	No	4	Bank Trans	Every 3 Mo	46.48
1750	43	Male	Kapeng Pa	Quezon Cit	Fall	3.2	No	No	No	36	Debit Card	Every 3 Mo	39.2
3606	68	Female	Kapeng Be	Baguio	Fall	4.3	No	No	No	36	Bank Trans	Every 3 Mo	22.96
1097	57	Male	Kapeng Am	Quezon Cit	Fall	2.6	No	Yes	Yes	48	Credit Card	Annually	25.76
2601	21	Male	Barako Co	Quezon Cit	Fall	3.9	No	No	No	6	Bank Trans	Bi-Weekly	46.48
1660	53	Male	Kapeng Be	Baguio	Fall	3.9	No	Yes	Yes	44	Venmo	Annually	20.72
3888	40	Female	Kapeng Am	Baguio	Spring	2.7	No	No	No	1	Credit Card	Quarterly	19.04
3498	35	Female	Kapeng Pa	Davao	Spring	4.3	No	No	No	16	Venmo	Every 3 Mo	26.88
3653	27	Female	Kapeng Ala	Quezon Cit	Fall	3.8	No	No	No	23	Debit Card	Monthly	16.24
1537	18	Male	Barako Co	Manila	Summer	2.6	No	Yes	Yes	17	Cash	Quarterly	45.92
2737	44	Female	Kapeng Pa	Quezon Cit	Spring	4.8	No	No	No	45	Credit Card	Annually	14
15	64	Male	Kapeng Pa	Cebu	Winter	4.7	Yes	Yes	Yes	34	Debit Card	Weekly	29.68
1026	60	Male	Barako Co	Davao	Winter	4.9	Yes	Yes	Yes	15	Credit Card	Annually	47.04
3255	21	Female	Kapeng Ba	Davao	Winter	4.9	No	No	No	11	Bank Trans	Every 3 Mo	17.36
2182	22	Male	Barako Co	Cebu	Fall	3.2	No	No	No	32	Bank Trans	Bi-Weekly	35.28
433	47	Male	Kapeng Ba	Cebu	Fall	3.3	Yes	Yes	Yes	26	Venmo	Weekly	48.16
2010	37	Male	Kapeng Ala	Quezon Cit	Summer	3.8	No	No	No	26	Venmo	Bi-Weekly	55.44
865	51	Male	Kapeng Be	Baguio	Winter	3.9	Yes	Yes	Yes	5	PayPal	Annually	33.04

coffee_customer_behavior_philip

c. Pre-processed Data



Previous Purchase	Review Rating	Discount Applied	Promo Code Used	Category_Barako Coffe	Category_Kapeng Alami	Category_Kapeng Amadi	Category_Kapeng Batan	Category_Kapeng Bengu	Category_Kapeng Cavite	Category_Kapeng Cordille	Category_Kapeng Davac	Category_Kapeng Palawa	Subscription Status
10	2.6	1	1	0	0	0	0	0	0	1	0	0	1
16	2.9	0	0	0	0	0	0	0	1	0	0	0	0
48	4.7	1	1	0	0	0	0	0	0	0	1	0	1
28	3.2	0	0	0	0	1	0	0	0	0	0	0	0
27	4.4	0	0	0	0	0	0	0	1	0	0	0	0
37	3.5	1	1	0	0	1	0	0	0	0	0	0	1
4	3.3	0	0	0	0	0	0	0	1	0	0	0	0
36	3.2	0	0	0	0	0	0	0	0	0	0	1	0
36	4.3	0	0	0	0	0	0	1	0	0	0	0	0
48	2.6	1	1	0	0	1	0	0	0	0	0	0	0
6	3.9	0	0	1	0	0	0	0	0	0	0	0	0
44	3.9	1	1	0	0	0	0	1	0	0	0	0	0
1	2.7	0	0	0	0	1	0	0	0	0	0	0	0
16	4.3	0	0	0	0	0	0	0	0	0	0	1	0
23	3.8	0	0	0	1	0	0	0	0	0	0	0	0
17	2.6	1	1	1	0	0	0	0	0	0	0	0	0
45	4.8	0	0	0	0	0	0	0	0	0	0	1	0
34	4.7	1	1	0	0	0	0	0	0	0	0	1	1
15	4.9	1	1	1	0	0	0	0	0	0	0	0	1
11	4.9	0	0	0	0	0	1	0	0	0	0	0	0
32	3.2	0	0	1	0	0	0	0	0	0	0	0	0
26	3.3	1	1	0	0	0	1	0	0	0	0	0	1
26	3.8	0	0	0	1	0	0	0	0	0	0	0	0
5	3.9	1	1	0	0	0	0	1	0	0	0	0	1
20	4.9	0	0	0	0	1	0	0	0	0	0	0	0
47	3	1	1	0	1	0	0	0	0	0	0	0	1
31	3.2	0	0	0	0	0	0	0	0	0	0	1	0
16	4	0	0	0	1	0	0	0	0	0	0	0	0
24	4.5	1	1	0	0	0	0	0	0	0	1	0	0
40	4.7	0	0	0	1	0	0	0	0	0	0	0	0
45	4.8	0	0	1	0	0	0	0	0	0	0	0	0
3	3.9	0	0	1	0	0	0	0	0	0	0	0	0
37	4.9	1	1	0	0	0	0	1	0	0	0	0	1
7	4.2	1	1	0	0	0	0	0	0	0	0	1	0
12	4.1	0	0	0	0	0	0	0	0	0	1	0	0
16	4.1	0	0	0	0	0	0	1	0	0	0	0	0
45	2.6	0	0	0	0	0	0	1	0	0	0	0	0
2	3.2	0	0	0	0	0	0	0	0	0	0	1	0
31	4.1	0	0	0	0	1	0	0	0	0	0	0	0
13	2.6	0	0	0	0	0	0	0	0	0	0	1	0
49	4	0	0	1	0	0	0	0	0	0	0	0	0
46	3.8	0	0	0	0	0	0	0	1	0	0	0	0
14	3.2	1	1	0	0	0	1	0	0	0	0	0	1

correlation | regression | customer_subscription_data | Sheet3 | Sheet1